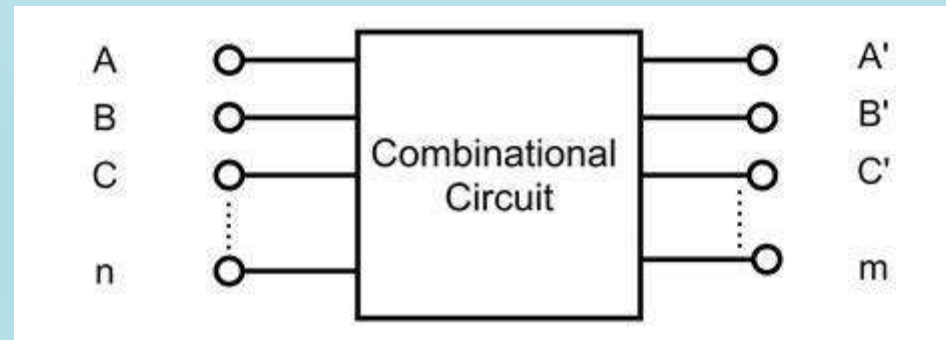


Combinational circuit

Combinational circuits are specially designed using multiple interconnected logic gates such that the output will be generated by computing the logical combinations of the **present input only**. No clock pulse is present here. Moreover, no previously stored value or state is taken into consideration here. **The output is independent of previous states.**

Example : Half/Full Adder, Half/Full Subtractor, Encoder/Decoder ,Multiplexer / Demultiplexer , etc .



Design Procedure of Combinational circuit

- 1) Find the required number of inputs and outputs and assign a symbol to each.
- 2) Derive the truth table according to given specifications and function.
- 3) Using the truth table, obtain simplified Boolean functions for each output as a function of the input variables.
- 4) Draw the logic circuit diagram.

Design a combinational circuit that has four input and two outputs. One of the outputs is high when majority of inputs are high . The second output is high only when all inputs are of same type.

HW: Advantage ,Disadvantage and application of Combinational circuits?

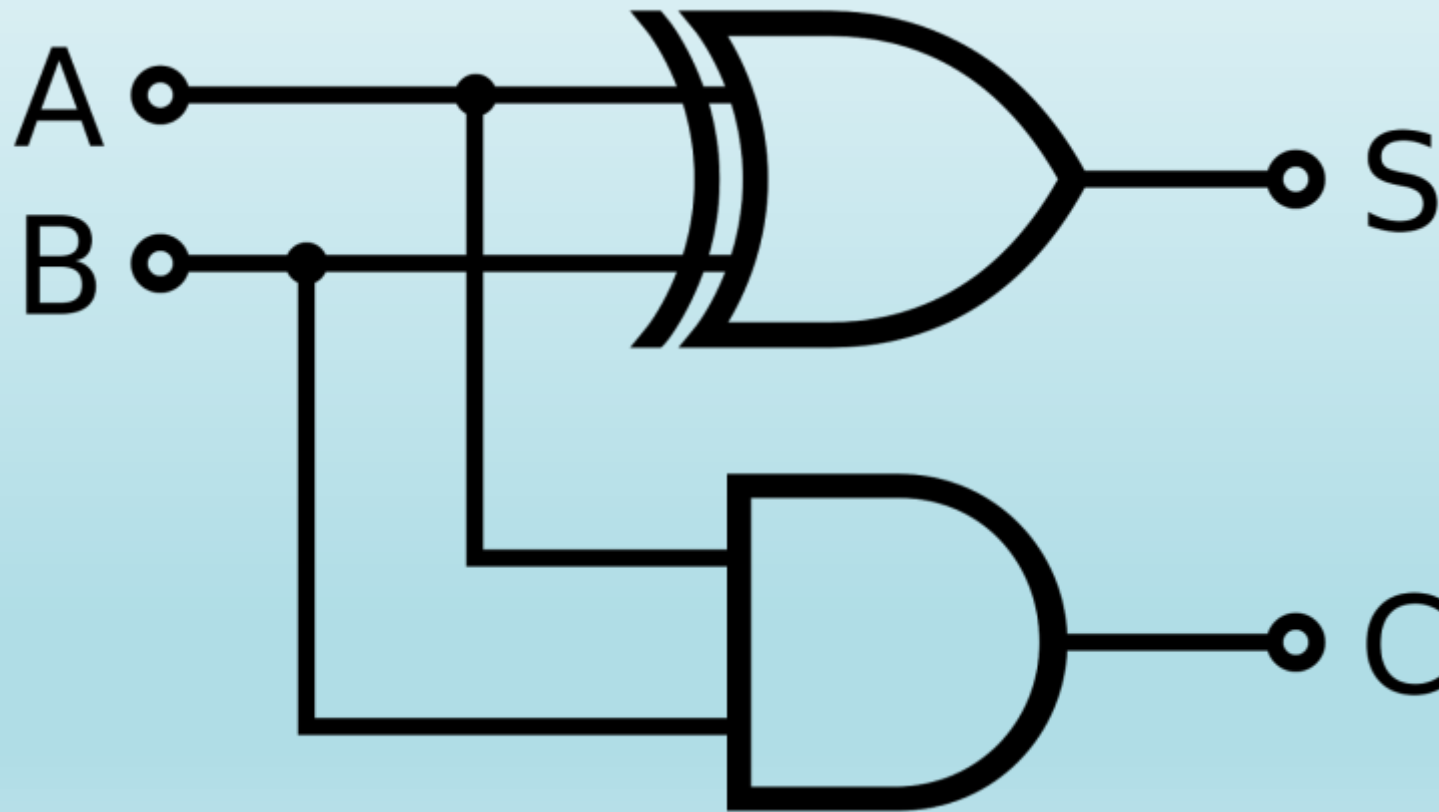
Half adder

The half adder is a basic building block for more complex adder circuits such as full adders and multiple-bit adders. It performs binary addition of two single-bit inputs, A and B, and provides two outputs, SUM and CARRY.

Truth table:

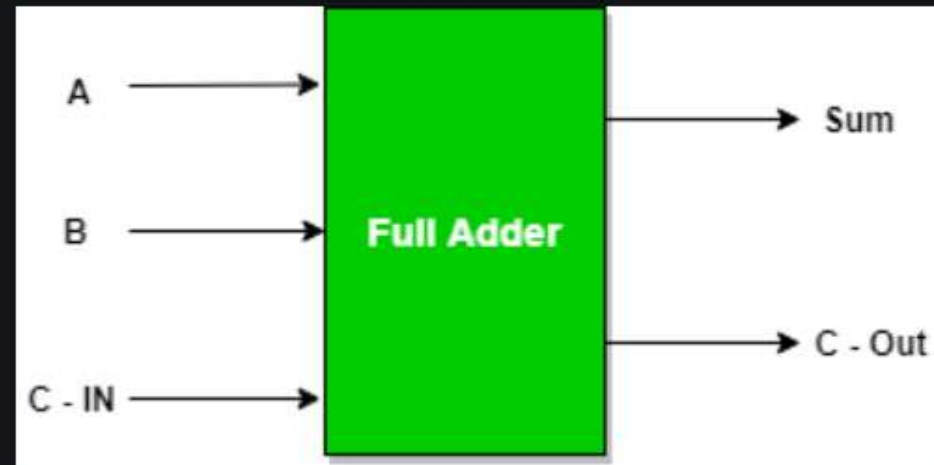
INPUTS		OUTPUTS	
A	B	Sum(S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Circuit diagram



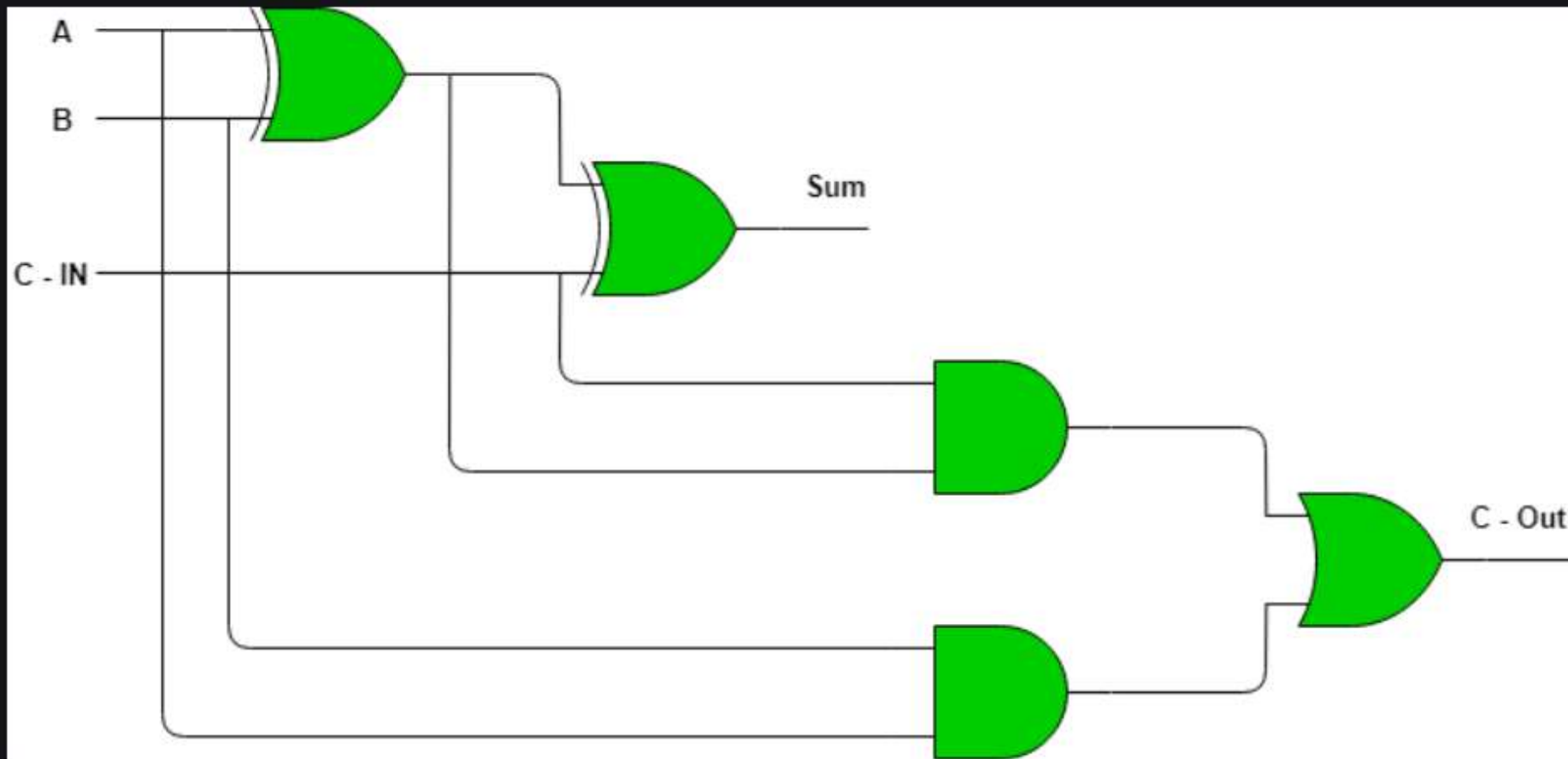
Full Adder

Full Adder is the adder that adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.



Full Adder Truth Table:

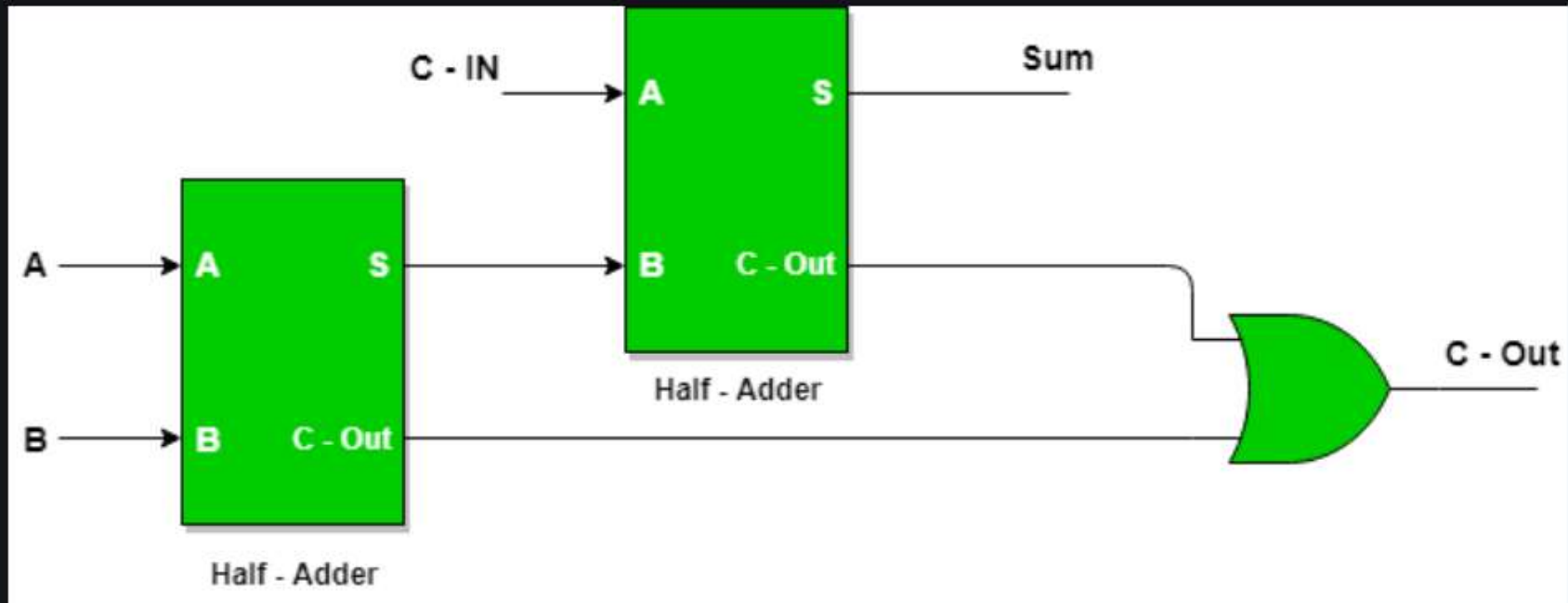
Inputs			Outputs	
A	B	C – IN	Sum	C - Out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Full Adder logic circuit.

Implementation of Full Adder using Half Adders

2 Half Adders and an OR gate is required to implement a Full Adder.



HW

Implementation of Full and Half adder using NAND gates and NOR gates.

Advantage ,Disadvantage and application of Half adder and Full adder.

Half Subtractor

Half subtractor is a combination circuit with two inputs and two outputs that are difference and borrow. It produces the difference between the two binary bits at the input and also produces an output (Borrow) to indicate if a 1 has been borrowed. In the subtraction $(A-B)$, A is called a Minuend bit and B is called a Subtrahend bit.



Half Subtractor

Truth Table

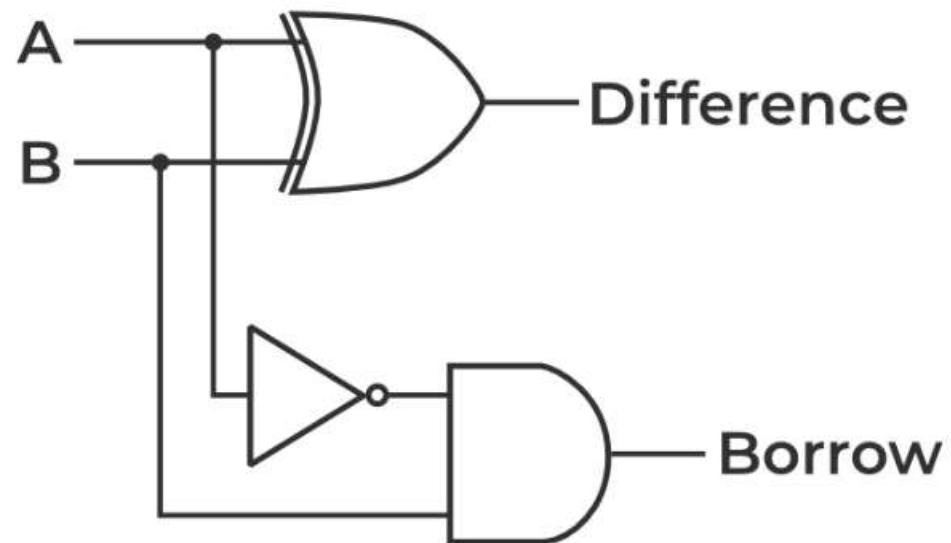
A	B	Diff	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

The SOP form of the Diff and Borrow is as follows:

$$\text{Diff} = A'B + AB'$$

$$\text{Borrow} = A'B$$

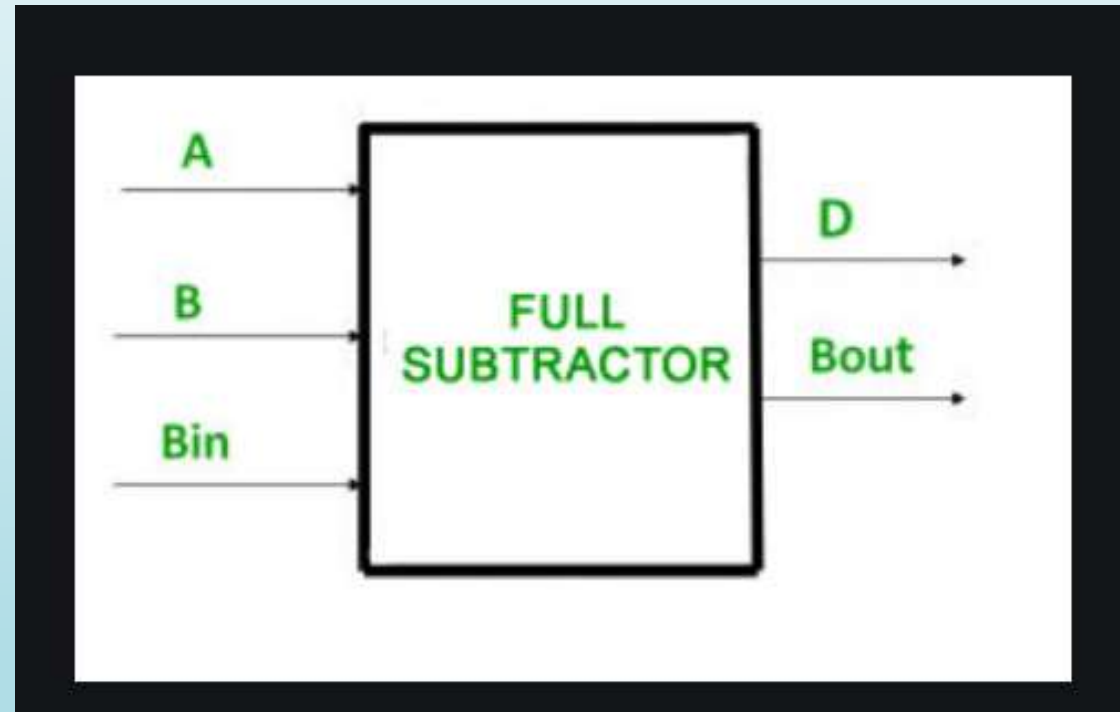
Implementation



Full Subtractor

A full subtractor is a combinational circuit that performs subtraction of two bits, one is minuend and other is subtrahend, taking into account borrow of the previous adjacent lower minuend bit. This circuit has three inputs and two outputs. The three inputs A, B and Bin, denote the minuend, subtrahend, and previous borrow, respectively. The two outputs, D and Bout represent the difference and output borrow.

Full Subtractor



Full Subtractor

Truth Table –

INPUT			OUTPUT	
A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

From above table we can draw the K-Map as shown for “difference” and “borrow”.

		B Bin			
		00	01	11	10
A	0	0	1	0	1
	1	1	0	1	0

$$D = A'B'Bin + AB'Bin' + A'BBin' + ABBin$$

		B Bin			
		00	01	11	10
A	0	0	1	1	1
	1	0	0	1	0

$$Bout = A'Bin + A'B + BBin$$

Full Subtractor

Logical expression for difference –

$$\begin{aligned} D &= A'B'Bin + A'BBin' + AB'Bin' + ABBin \\ &= Bin(A'B' + AB) + Bin'(AB' + A'B) \\ &= Bin(A \text{ XNOR } B) + Bin'(A \text{ XOR } B) \\ &= Bin(A \text{ XOR } B)' + Bin'(A \text{ XOR } B) \\ &= Bin \text{ XOR } (A \text{ XOR } B) \\ &= (A \text{ XOR } B) \text{ XOR } Bin \end{aligned}$$

Logical expression for borrow –

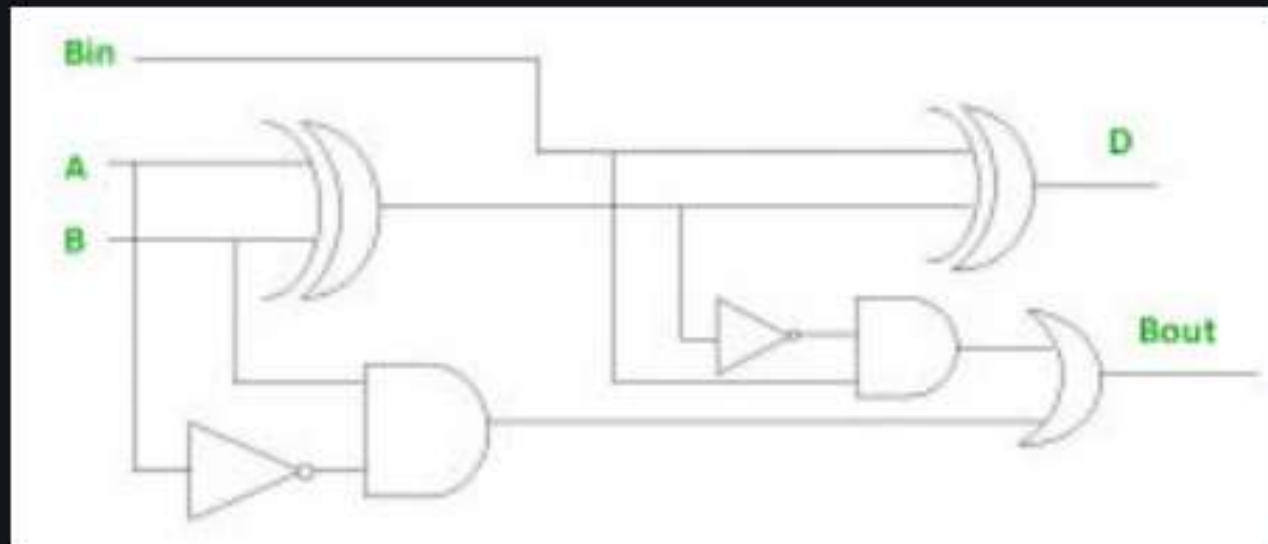
$$\begin{aligned} Bout &= A'B'Bin + A'BBin' + A'BBin + ABBin \\ &= A'B'Bin + A'BBin' + A'BBin + A'BBin + A'BBin + ABBin \\ &= A'Bin(B + B') + A'B(Bin + Bin') + BBin(A + A') \\ &= A'Bin + A'B + BBin \end{aligned}$$

OR

$$\begin{aligned} Bout &= A'B'Bin + A'BBin' + A'BBin + ABBin \\ &= Bin(AB + A'B') + A'B(Bin + Bin') \\ &= Bin(A \text{ XNOR } B) + A'B \\ &= Bin(A \text{ XOR } B)' + A'B \end{aligned}$$

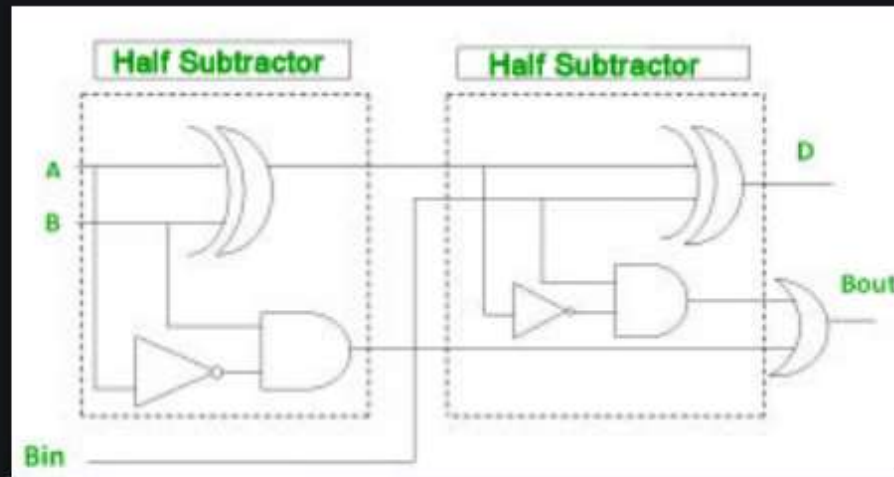
Full Subtractor

Logic Circuit for Full Subtractor –



Full Subtractor

Implementation of Full Subtractor using Half Subtractors – 2 Half Subtractors and an OR gate is required to implement a Full Subtractor.



HW

Advantage , Disadvantage and Application of Half subtractor and full subtractor.

Full Subtractor using NAND gates and NOR gates only.